

SOFTWARE REQUIREMENTS SPECIFICATION (SRS) FOR RESTAURANT APP MVP

1. Introduction

1.1 Purpose: This document defines the requirements of the Restaurant App MVP, its intended audience (Diners and Managers), and the frontend-only scope for Fall 2026.

1.2 Scope: In scope: Authentication, Pakistani Menu browsing, search with debouncing, global themes, cart management (`useReducer`), performance optimizations (`useMemo`), custom hooks, table reservation, and Manager dashboard. Out of scope: Real online payments, push notifications, and backend server database.

1.3 Definitions and Acronyms

SRS: Software Requirements Specification

MVP: Minimum Viable Product

UML: Unified Modeling Language

Hook: React function for state and side effects

Context: React API for global state sharing

Reducer: Pure function managing complex state transitions

Mock Data: Local JSON/JS data sets simulating backend

FlatList: High-performance scrolling list component in React Native

2. Overall Description

2.1 User Roles:

Customer: Browse menu, manage cart, apply promo codes, book tables, track orders.

Manager: Manage incoming orders, accept/decline table reservations, update menu item availability and prices.

2.2 User Stories:

As a customer, I want to browse Pakistani dishes so that I can order food quickly.

As a customer, I want to filter items by category so that I can find my favorite Biryani or Karahi.

As a customer, I want to apply promo codes like `WELCOME10` so that I get a discount.

As a customer, I want to book a table for 7:00 PM so that my seating is ready.

As a customer, I want to view my order tracking status in real-time.

As a manager, I want to view incoming orders so that I can process them efficiently.

As a manager, I want to update order statuses (Pending -> Preparing -> Ready -> Served) so that customers stay updated.

As a manager, I want to toggle menu item availability so that unavailable dishes are hidden.

3. Functional Requirements

FR-01: The system shall allow users to log in or sign up with role selection (Customer/Manager).

FR-02: The system shall display a Pakistani menu categorized into Starters, Mains, Desserts, and Drinks.

FR-03: The system shall provide a debounced search bar to filter menu items.

FR-04: The system shall manage shopping cart state using `useReducer` supporting item increments, decrements, and notes.

FR-05: The system shall allow table reservations with Pakistani phone number validation (`03XX-XXXXXXX`).

4. Non-Functional Requirements

Usability: Simple, intuitive interface tailored for Pakistani users.

Performance: Optimized with `useMemo`, `useCallback`, and `React.memo` to prevent unnecessary re-renders.

Responsiveness: Adapts seamlessly across various mobile screen sizes.

5. Client-Side Data Model

Stored locally in `src/data/` as JavaScript mock arrays for Users, Categories, MenuItems, Tables, Reservations, and Orders.

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